# KARTHIK MARIMUTHU

**MLOps Engineer | ML, GenAI & LLM Platforms | Royal Bank of Canada**

Toronto, Canada | Open to relocation to UAE, available on 2 weeks' notice

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github.com/KarthikTools | karthiktools.github.io

## SUMMARY

Platform engineer specializing in MLOps, with 12+ years in financial services taking machine learning, generative AI (GenAI), and large language model (LLM) systems from pilot to governed production. At Royal Bank of Canada, architected and lead the serving and release platform for the bank's first production agentic AI product: LLM serving on Kubernetes/OpenShift, rollback-safe CI/CD release automation, runtime observability, and the safety controls that cleared the bank's architecture, security, and governance reviews. The classical ML lifecycle is demonstrated in four public, CI-verified reference platforms (github.com/KarthikTools): end-to-end ML pipelines covering model training, validation, and deployment, MLflow registries, challenger-vs-champion promotion gates, Evidently drift detection with automated retraining, and evaluation-gated LLM releases.

## SKILLS

**MLOps and model lifecycle:** MLflow (experiment tracking, model registry, aliases), model lineage, champion/challenger promotion gates, drift detection and model monitoring (Evidently), continuous training (CT), LLM evaluation gates, prompt registries, agent lifecycle operations (AgentOps), model deployment and rollback (demonstrated in the public reference platforms below)

**CI/CD and orchestration:** CI/CD (continuous integration/continuous delivery) pipelines, GitHub Actions, Bamboo, Bitbucket, Airflow, release automation, canary rollouts with automatic rollback

**Model serving and infrastructure:** FastAPI model serving APIs, real-time and batch inference, Kubernetes, OpenShift, Docker, autoscaling, health probes, Terraform infrastructure as code

**Cloud:** Amazon Web Services (AWS): EKS, Lambda, S3, IAM, CloudWatch, SageMaker (provisioning and network/KMS hardening via Terraform); Microsoft Azure; VPC isolation, private endpoints, KMS encryption

**Data and AI systems:** Python, SQL, Spark (PySpark), PostgreSQL, vector databases (pgvector), Redis, data ingestion and ETL/ELT pipelines, retrieval-augmented generation (RAG), natural language processing (NLP), transformer-based LLM systems, scikit-learn, LangGraph, LangChain, Model Context Protocol (MCP), Java, Spring Boot

**Governance and security:** model governance and audit trails, human-in-the-loop (HITL) approval controls, Responsible AI guardrails (PII redaction, prompt-injection defense), risk-proportionate agent controls, role-based access control (RBAC), least-privilege IAM; reference-platform implementations designed against OSFI E-23, SR 11-7, and CBUAE model risk management principles

## WORK EXPERIENCE

### Royal Bank of Canada (RBC), Toronto

***Lead Platform Engineer, AI Transformation*** | May 2022 - Present

- Architected and lead the serving and release platform for RBC's first production agentic AI product, adopted by multiple business units: FastAPI-based LLM and agent services on OpenShift (OCP4) Kubernetes with health probes, autoscaling, shared base-service templates, and rollback-safe release automation.
- Took the platform through the bank's architecture, security, and governance reviews by building the guardrail layer: PII redaction, prompt-injection defense, graph-level RBAC, human-in-the-loop approval for agent code execution, and request tracing with structured logging of model interactions.
- Wrote the observability standard for request tracing, model runtime latency tracking, structured logging, and CI/CD pipeline metrics, now followed across the platform's AI microservices.
- Built the data ingestion and ETL pipeline that turns earnings-call transcripts from the six largest Canadian banks into a governed, searchable vector database (PostgreSQL/pgvector), powering a citation-grounded retrieval-augmented generation (RAG) research agent with fail-closed data governance controls on regulated policy documents.
- Designed the platform's extension model so new domains, agent skills, and MCP tool servers ship as configuration with no core code changes; new AI use cases now onboard through config review instead of platform engineering work.
- Mentored 8 engineers on agent architecture and cloud-native patterns; defended AI system designs in front of architecture, security, and executive stakeholders. Spot Award for enterprise AI transformation.

### Virtusa Consulting Services, Canada

***Senior DevOps Engineer*** | Aug 2016 - Apr 2022

- Operated a distributed wealth-management microservices platform processing ~10M transactions a day; held its 99.9% availability target through capacity planning and fault-tolerance design.

- Rebuilt CI/CD pipelines on Bitbucket, Bamboo, OpenShift, and Docker, cutting deployment time by 40%; Star Award for automation work.
- Built production Python ML pipelines for financial document validation and comparison, improving validation accuracy by 45% in daily document workflows.
- Led a Technical Center of Excellence producing shared Python/FastAPI libraries and automation frameworks; designed cloud-native systems on AWS and Microsoft Azure serving up to 50K concurrent users.

### HCL Technologies, India

***Software Developer*** | Nov 2014 - Jul 2016

- Refactored insurance-domain monoliths into microservices (Java, Spring Boot, Python), cutting legacy processing time by 25% through SQL/NoSQL optimization.

## OPEN-SOURCE WORK: REGULATED-ML REFERENCE PLATFORMS

Four public, CI-verified platforms implementing the full model lifecycle with regulated-banking control patterns (github.com/KarthikTools):

- **credit-risk-mlops-platform**: trains and compares supervised learning candidates (scikit-learn gradient boosting, logistic regression) with a PySpark feature-engineering stage (window and aggregate features); MLflow model registry with promotion gated on challenger-beats-champion evaluation, FastAPI serving, and Evidently drift detection that triggers continuous training with tracked retraining events; CI/CD/CT pipelines with a manual production approval gate.
- **model-drift-monitor**: drift statistics exposed as Prometheus metrics, Grafana dashboards, and graduated alerting for silent model failure.
- **terraform-ml-platform-aws**: VPC-isolated SageMaker and private EKS with customer-managed KMS keys, least-privilege IAM, and Checkov policy-as-code scanning in CI.
- **llmops-eval-pipeline**: versioned prompt registry, evaluation gates that block quality regressions in CI, and canary rollout with automatic rollback.

Roadmap: Databricks port of the Spark feature stage; Feast feature-store extension; Azure ML deployment port. Technical writing on CI/CD/CT pipeline design and production drift monitoring: karthiktools.github.io

## EDUCATION

B.E. Electronics and Communication Engineering, Dr. Mahalingam College of Engineering and Technology, 2014
Deep Learning Specialization (Coursera)